SEA4001W: Exercise 9 - Forward in time, centred in space (FCTS)

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Exercise 9

eq.29
$$\frac{\partial^2 U}{\partial x^2} = \frac{U(x+\Delta x) - 2U(x) + U(x_i - \Delta x)}{\Delta x^2} + O(\Delta x)$$

$t^n = t_0 + n\Delta t$

$x_i = x_0 + i\Delta x$

$$\begin{cases} U(x, t_n) = U_i^n \\ U(x+\Delta x, t_n) = U_{i+1}^n \\ U(x-\Delta x, t_n) = U_{i-1}^n \end{cases}$$

moving one spatial step forward means going from index i to $i+1$

one step backward means going $i-1$

take at the same time level n

$$\frac{\partial^2 U}{\partial x^2} \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{\Delta x^2}$$

$$\frac{\partial U}{\partial z} = \frac{U(z_i+\Delta z) - U(z_i - \Delta z)}{2\Delta z} + O(\Delta z^2)$$

$U_{ijk}^n = U(x_i, y_j, z_k, t_n)$

i for x, j for y, k for z, & n for time

i, j, n stay fixed

index k changes

$z_k = z_0 + k\Delta z$

$U(z_k + \Delta z) = U_{i,j,k+1}^n$ & $U(z_k - \Delta z) = U_{i,j,k-1}^n$

moving one step forward in z increases k index by 1, & moving one step back decreases it by 1

$$\frac{\partial U}{\partial z} \approx \frac{U_{i,j,k+1}^n - U_{i,j,k-1}^n}{2\Delta z}$$